



RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE
(Autonomous Institute, Affiliated to VTU)
DEPARTMENT OF CIVIL ENGINEERING

Term: June to Sept 2023 **Course:** Introduction to Civil Engineering **Code:** ESC231

Max Marks: 30

Duration: 60 min

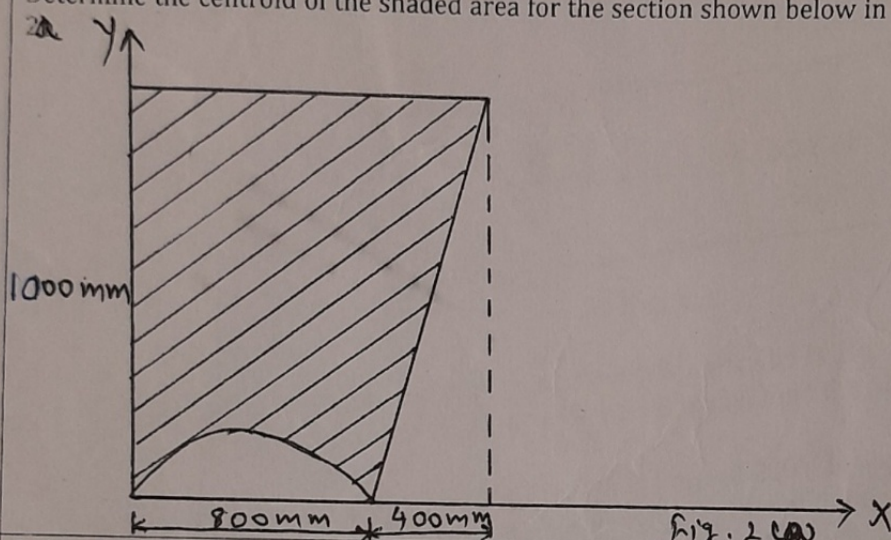
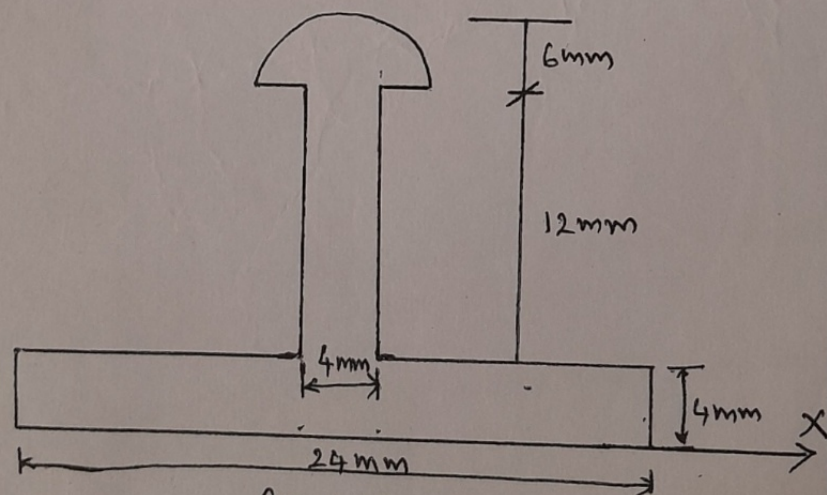
CIE: Test II

Sem: 2nd

Sec: --

Instructions to Candidates:

Answer any **TWO** full questions of the followings.

Sl#	Question	CO	BL	Marks
1 (a)	Discuss in brief the societal and global impact of infrastructure.	2	2	07
1 (b)	State and prove parallel axis theorem.	5	3	08
2(a)	Determine the centroid of the shaded area for the section shown below in figure 2a  Fig. 2(a)	4	3	08
2(b)	Define friction. State the various laws of static and dynamic friction.	4	3	07
3(a)	Determine the Moment of Inertia about its horizontal axis for the section shown below in figure 3a  Fig. 3(a)	5	3	08
3(b)	Define Centroid. Determine the centroid of rectangular lamina.	4	2	07